



AGENTIC WORKOUT PLANNER



Master of Science in
Artificial Intelligence

INTERPRETABLE, INJURY-AWARE PLANNING USING SYMBOLIC REASONING
AND GENERATIVE AI.

PROBLEM

- Safety Failure:** Current fitness AI (LLMs and templates) cannot guarantee user safety because they lack deterministic rules to prevent injury.
- Personalization Limits:** Standard systems fail to adapt to individual factors such as joint–muscle mechanics and equipment availability.
- Transparency Problem:** Opaque AI models provide recommendations without explaining why exercises are allowed or restricted.

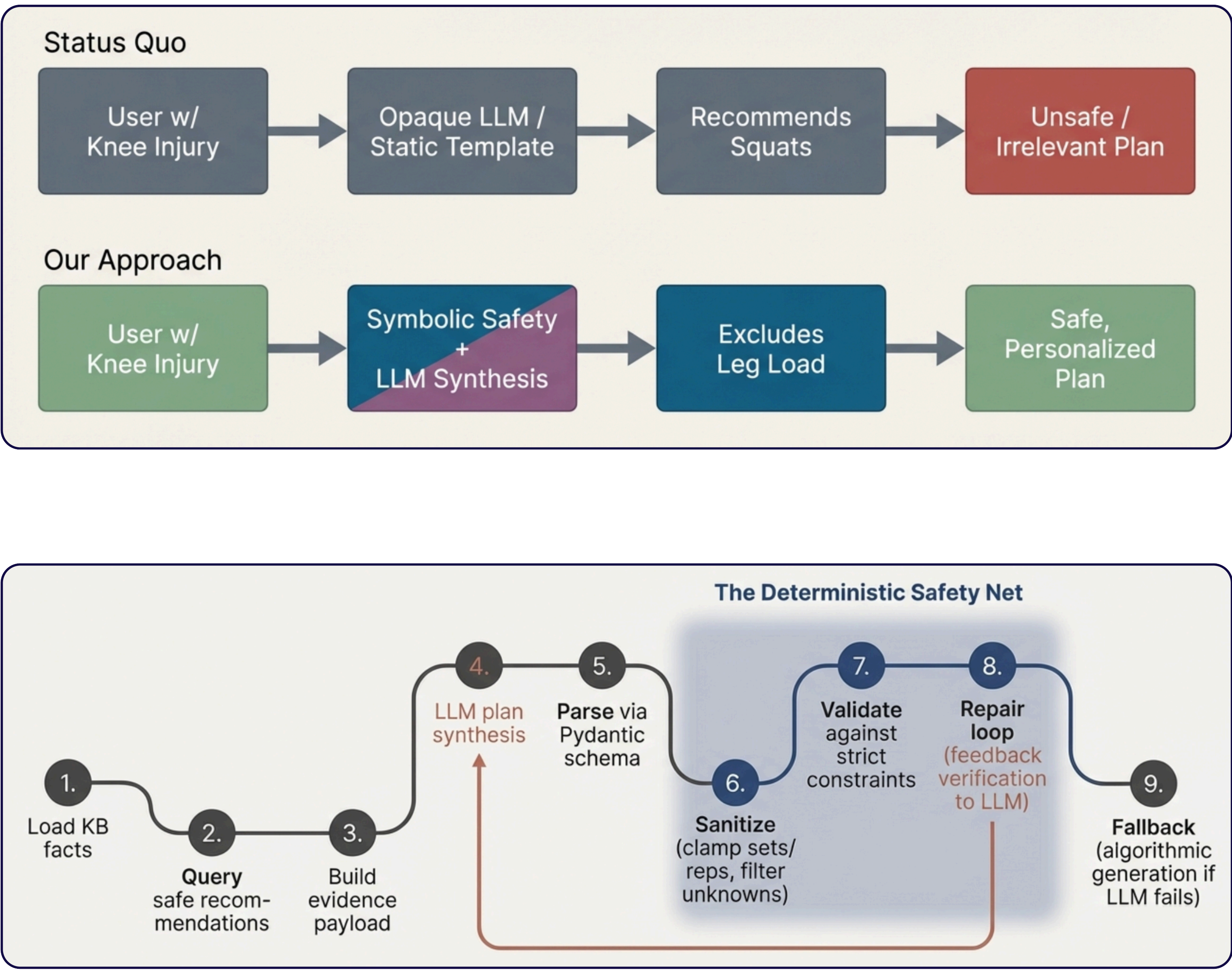
WHY IT MATTERS

- Value Proposition:** Combining artificial intelligence with symbolic logic prevents unsafe, hallucinated workout routines by enforcing deterministic safety constraints.
- Who Benefits:** Users with injuries, limited equipment, or specialized fitness needs that require reliable, personalized workout guidance.
- Our Solution:** Replaces unpredictable AI routines with verifiable plans and interpretable reasoning that explains why exercises are recommended or restricted.

COMPONENTS

- ExerciseDB:** Structured exercise database with muscle–joint, exercise–equipment, and injury–muscle relationships.
- Rule-based logic engine:** Equipment + Injury focused constraint-based evaluation for exercise selection.
- Agentic Orchestration:** Autonomous updates to the plan based on changes in injury, intensity, and equipment.
- Clean, interactive UI:** Intuitive, chat-based interface for users to interact with the agent and keep track of their information.

METHODOLOGY



REASONING

Facts

- is_exercise(E)
- trains_muscle(E, M)
- uses_equipment(E, Eq)
- injured_muscle(User, M)
- preferred_equipment(User, Eq)

Safety Rules

- forbidden(User, E) ← trains_muscle(E, M) ∧ injured_muscle(User, M)
- equipment_allowed(User, E) ← preferred_equipment(U, Eq) ∧ uses_equipment(E, Eq)

Goal

- recommended(User, E) ← is_exercise(E) ∧ ¬forbidden(User, E) ∧ equipment_allowed(User, E)

